# Type MS · Define a reflex and name the neurones on the receptor-to-effector path

## Answer 1 of 13 · Question 5 part b · 3 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

| Number | | | |
|---|---|---|---|
| 5 (a) | any three from<br>• rapid response (to a stimulus) (1)<br>• involuntary/automatic (1)<br>• no brain involvement (1)<br>• protective (1) | | 3 max |
| (b) | • A = motor (1)<br>• B = relay/association/connector (1)<br>• C = sensory (1) | | 3 |
| (c) (i) | • fewer branches/dendrons/dendrites (1)<br>• shorter branches/ dendrons/dendrites (1)<br>• fewer knobs/synapses (1) | R axons | 3 |
| (ii) | • transmits impulse (1)<br>• across synapse/(synaptic) cleft (1)<br>• stimulates impulse in next neurone (1) | | 3 |
| (iii) | • secreted by synapses/synaptic knob (1)<br>• fewer synapses/knobs (1)<br>• less secreted by remaining synapses/knobs (1) | | 3 |

Total 15 marks

## Type MS · Define a reflex and name the neurones on the receptor-to-effector path

### Answer 1 of 13 · Question 5 part b · 3 marks

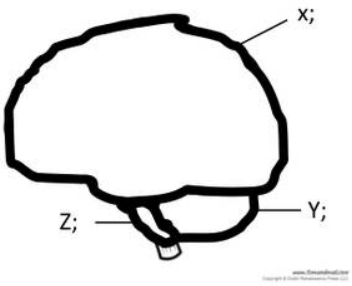
2020 NOV · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 2 of 13 · Question 7 part b.i · 1 mark

2019 JUN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 7 (a) |  | Letters can be anywhere within outline shown.<br>Allow labels in place of than letters. | 3 |
| (b) (i) | Motor; | | 1 |
| (ii) | Any one from the following:<br><br>(sensory neurone):<br>longer dendrites/shorter axon/cell body outside of spinal cord/CNS/in middle of neurone/transmit impulse to the CNS/brain/spinal cord<br><br>(motor neurone)<br>shorter dendrites/longer axon/cell body at one end of neurone/inside spinal cord/transmit impulses from CNS/brain/spinal cord | Answer must be comparative and state which type of neurone is being referred to | 1 |
| (iii) | • chemicals/neurotransmitters;<br>• diffuse;<br>• across a synapse;<br>• bind to receptors (on post-synaptic membrane); | | 3 |
| (c) | • stem cells are undifferentiated/unspecialised/can differentiate;<br>• could be made to form nerve cells;<br>• to replace damaged cells in brain; | | 1<br>1<br>1 |

Total question 7 = 11 marks

## Type MS · Define a reflex and name the neurones on the receptor-to-effector path

### Answer 2 of 13 · Question 7 part b.i · 1 mark

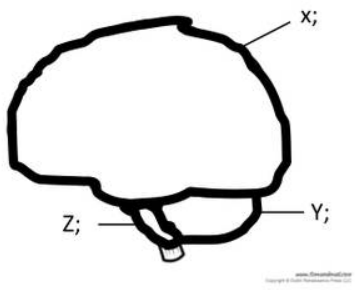
2019 JUN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 3 of 13 · Question 7 part b.ii · 1 mark

2019 JUN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 7 (a) |  | Letters can be anywhere within outline shown.<br>Allow labels in place of than letters. | 3 |
| (b) (i) | Motor; | | 1 |
| (ii) | Any one from the following:<br><br>(sensory neurone):<br>longer dendrites/shorter axon/cell body outside of spinal cord/CNS/in middle of neurone/transmit impulse to the CNS/brain/spinal cord<br><br>(motor neurone)<br>shorter dendrites/longer axon/cell body at one end of neurone/inside spinal cord/transmit impulses from CNS/brain/spinal cord | Answer must be comparative and state which type of neurone is being referred to | 1 |
| (iii) | • chemicals/neurotransmitters;<br>• diffuse;<br>• across a synapse;<br>• bind to receptors (on post-synaptic membrane); | | 3 |
| (c) | • stem cells are undifferentiated/unspecialised/can differentiate;<br>• could be made to form nerve cells;<br>• to replace damaged cells in brain; | | 1<br>1<br>1 |

Total question 7 = 11 marks

## Type MS · Define a reflex and name the neurones on the receptor-to-effector path

### Answer 3 of 13 · Question 7 part b.ii · 1 mark

2019 JUN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 4 of 13 · Question 3 part b · 1 mark

2021 NOV · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

| number | | | |
|---|---|---|---|
| 3 (a) | C hypothalamus;<br><br>A does not control sweating<br>B does not control sweating<br>D does not control sweating | | 1 |
| (b) | B motor;<br><br>A transmits impulse from sensory to motor<br>C alternative name for A with same function<br>D transmits impulse to brain | | 1 |
| (c) | • (sweat) evaporates from (the surface of) the skin;<br>• using heat from body/blood;<br>• cools body/reduces body temperature; | | 3 |
| (d) | • runs in families;<br>• must be faulty/mutated gene/chromosome/allele;<br>• which transfers/passes on information to offspring; | | 3 |
| (e) | obesity / high sugar intake / high BMI; | | 1 |
| (f) | • exercise/activity of individual varies;<br>• greater rate of aerobic respiration;<br>• heat given off as by-product/increased body temperature;<br>• external temperature varies;<br>• hotter more sweating/cooler less sweating;<br>• mass of individual varies;<br>• volume of water in the body/fluid intake varies; | Allow reverse argument for mps 2 and 3 | 5<br><br>Total 14 |

## Type MS · Define a reflex and name the neurones on the receptor-to-effector path

### Answer 4 of 13 · Question 3 part b · 1 mark

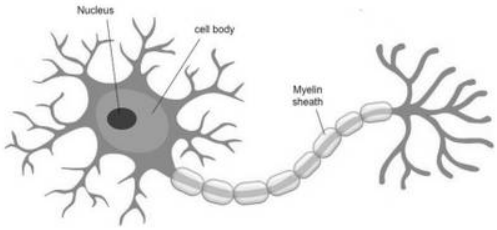
2021 NOV · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 5 of 13 · Question 7 part a.ii · 1 mark

2024 JUN · Paper 01 (regional R) · find part a.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 7 (a) (i) | receptor (cell) | | (1) |
| (ii) | sensory neurone | | (1) |
| (b) (i) | a cell that has particular features/adaptations/structures;<br>to carry out a certain/specific function in the body; | | (2) |
| (ii) | diagram; to include cell body at one end, with dendrites and an elongated axon twice the length of the cell body and axon ending;<br>nucleus;<br>cell body;<br>myelin sheath;<br>  <br> | Label marks can be awarded if they draw a sensory or relay neurone | (4) |
| (iii) | transmits electrical impulses from the brain/CNS/spinal cord/ relay neurone;<br>to effecter organs/muscle/gland; | | (2) |
| (c) | 100/4.0 x $10^{6}$;<br>= 0.000025 (s);<br>= 0.025 (ms);<br>= 2.5 x $10^{-2}$; | ecf for mps 2,3 and 4<br><br>Allow full marks for correct final answer | (4) |

**Total for Question 7 = 14 marks**

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 6 of 13 · Question 1 part a.i · 1 mark

2024 JUN · Paper 01 · find part a.i in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1 (a) | (i) | sensory (neurone/cell); | | (1) |
| | (ii) | axon; | | (1) |
| (b) | (i) | Part R sweat gland;<br>Part S blood vessels/capillaries/arterioles; | | (2) |
| | (ii) | Any two from:<br>• produces sweat/perspiration;<br>• sweat evaporates (from skins surface);<br>• using heat energy (from the blood); | | (2) |
| | (iii) | • provides an insulating layer insulation;<br>• traps heat (energy)/to prevent heat loss; | reject keeps body warm | (2) |
| (c) | (i) | • water is too hot/burn hands; | | (1) |
| | (ii) | • use a thermometer;<br>• to test the temperature of the water/to make sure the water is not too hot; | | (2) |
| | (iii) | • (temperature) receptor; | | (1) |

**Total for Question 1 = 12 Marks**

## Type MS · Define a reflex and name the neurones on the receptor-to-effector path

### Answer 7 of 13 · Question 3 part b.i · 3 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 3 | (a) | | • rapid/fast/eq. ;<br>• involuntary/automatic action;<br>• does not involve conscious thought;<br>• protective/prevents (further) damage; | allow instant-mp1<br>ignore ref to brain<br>ignore sub-conscious<br>ignore unconscious | 3 |
| | (b) | (i) | • sensory;<br>• relay/intermediate;<br>• motor; | correct positions on diagram | 3 |
| | | (ii) | correct direction from sensory to motor; | Mark correctly if there are more than 1 arrow, unle/wrongss they are in opposite direction | 1 |
| | | (iii) | synapse/synaptic cleft; | | 1 |
| | | | | | **Total = 8** |

## Type MS · Define a reflex and name the neurones on the receptor-to-effector path

### Answer 7 of 13 · Question 3 part b.i · 3 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 8 of 13 · Question 3 part a · 4 marks

2022 JUN · Paper 02 · find part a in this mark scheme extract

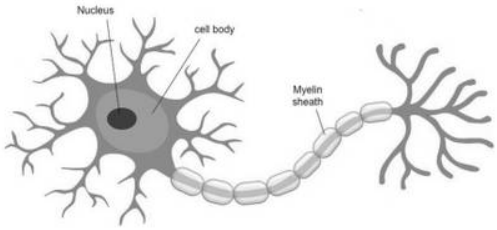
| | | | |
|---|---|---|---|
| 3 (a) | In the following order:<br>• receptors;<br>• sensory;<br>• relay;<br>• motor; | | 4 |
| (b) (i) | synapse; | | 1 |
| (ii) | • chemicals/neurotransmitters;<br>• diffuse across synaptic clef/gap; | | 2 |
| (c) | • 60 × 5;<br>• 300 x 120 = 36 000; | | 2 |
| | | | Total 9 |

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Define a reflex and name the neurones on the receptor-to-effector path

Answer 9 of 13 · Question 7 part b.iii · 2 marks

2024 JUN · Paper 01 (regional R) · find part b.iii in this mark scheme extract

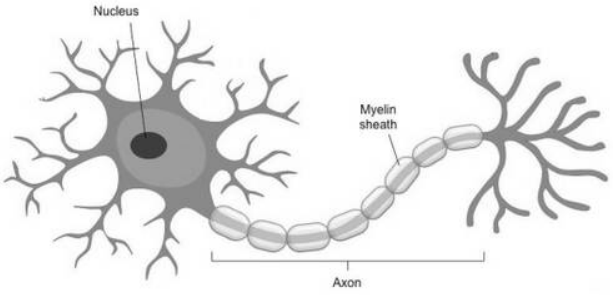
| | | | |
|---|---|---|---|
| 7 (a) (i) | receptor (cell) | | (1) |
| (ii) | sensory neurone | | (1) |
| (b) (i) | a cell that has particular features/adaptations/structures;<br>to carry out a certain/specific function in the body; | | (2) |
| (ii) | diagram; to include cell body at one end, with dendrites and an elongated axon twice the length of the cell body and axon ending;<br>nucleus;<br>cell body;<br>myelin sheath;<br>  <br> | Label marks can be awarded if they draw a sensory or relay neurone | (4) |
| (iii) | transmits electrical impulses from the brain/CNS/spinal cord/ relay neurone;<br>to effecter organs/muscle/gland; | | (2) |
| (c) | 100/4.0 x $10^{6}$;<br>= 0.000025 (s);<br>= 0.025 (ms);<br>= 2.5 x $10^{-2}$; | ecf for mps 2,3 and 4<br><br>Allow full marks for correct final answer | (4) |

**Total for Question 7 = 14 marks**

## Type MS · Define a reflex and name the neurones on the receptor-to-effector path

### Answer 10 of 13 · Question 3 part a.ii · 2 marks

2025 JUN · Paper 02 (regional R) · find part a.ii in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 3 | (a) | (i) |  | 1 mark for diagram 3 marks for labels | 4 |
| | | (ii) | • transmits electrical/nerve impulses;<br>• from CNS/brain/spinal cord/relay neurone, to effector organs/named effector organ; | | 2 |
| | (b) | (i) | Any six from<br>1 (student X) holds up the ruler between the thumb and forefinger (of other student);<br>2 then drops/releases the ruler;<br>3 (other) student catches the ruler;<br>4 record the reading on the ruler/distance fallen when caught;<br>5 repeat for <u>this</u> student;<br>6 using the same distance between thumb and forefinger;<br>7 find mean/average value of reading on ruler;<br>8 calculate reaction time (using formula);<br>9 repeat with <u>other</u> students; | | 6 |
| | | (ii) | the reading on the ruler/ reaction time; | | 1 |
| | | (iii) | independent variable not continuous/is discrete; | | 1 |
| | | (iv) | reaction time = √(0.15 ÷ 5);<br>reaction time = 0.17 (s); | accept 0.173(2) | 2 |
| | | | | **Total for Question 3 = 16 marks** | |

## Type MS · Define a reflex and name the neurones on the receptor-to-effector path

### Answer 11 of 13 · Question 5 part a · 3 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

| Number | | | |
|---|---|---|---|
| 5 (a) | any three from<br>• rapid response (to a stimulus) (1)<br>• involuntary/automatic (1)<br>• no brain involvement (1)<br>• protective (1) | | 3 max |
| (b) | • A = motor (1)<br>• B = relay/association/connector (1)<br>• C = sensory (1) | | 3 |
| (c) (i) | • fewer branches/dendrons/dendrites (1)<br>• shorter branches/ dendrons/dendrites (1)<br>• fewer knobs/synapses (1) | R axons | 3 |
| (ii) | • transmits impulse (1)<br>• across synapse/(synaptic) cleft (1)<br>• stimulates impulse in next neurone (1) | | 3 |
| (iii) | • secreted by synapses/synaptic knob (1)<br>• fewer synapses/knobs (1)<br>• less secreted by remaining synapses/knobs (1) | | 3 |

Total 15 marks

## Type MS · Define a reflex and name the neurones on the receptor-to-effector path

### Answer 11 of 13 · Question 5 part a · 3 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

## Type MS · Define a reflex and name the neurones on the receptor-to-effector path

### Answer 12 of 13 · Question 8 part a · 3 marks

2021 JUN · Paper 01 · find part a in this mark scheme extract

| number | | | | |
|---|---|---|---|---|
| 8 | (a) | any three from<br>• rapid response;<br>• automatic;<br>• to protect body;<br>• from harmful effects/named example; | | 3 |
| | (b) | correct position; | | 1 |
| | (c) | any six from<br>• impulse arrives at end of axon;<br>• synapse;<br>• causes release of neurotransmitter;<br>• e.g. acetylcholine;<br>• from vesicles;<br>• diffuses across gap/cleft;<br>• combines with/stimulates (post synaptic) receptors;<br>• triggers impulse; | | 6 |

Total 10 marks

## Type MS · Define a reflex and name the neurones on the receptor-to-effector path

### Answer 13 of 13 · Question 3 part a · 3 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 3 | (a) | | • rapid/fast/eq. ;<br>• involuntary/automatic action;<br>• does not involve conscious thought;<br>• protective/prevents (further) damage; | allow instant-mp1<br>ignore ref to brain<br>ignore sub-conscious<br>ignore unconscious | 3 |
| | (b) | (i) | • sensory;<br>• relay/intermediate;<br>• motor; | correct positions on diagram | 3 |
| | | (ii) | correct direction from sensory to motor; | Mark correctly if there are more than 1 arrow, unle/wrongss they are in opposite direction | 1 |
| | | (iii) | synapse/synaptic cleft; | | 1 |
| | | | | | **Total = 8** |

## Type MS · Define a reflex and name the neurones on the receptor-to-effector path

### Answer 13 of 13 · Question 3 part a · 3 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |